

name: RDP

on:

workflow_dispatch:

jobs:

secure-rdp:

runs-on: windows-latest

timeout-minutes: 3600

steps:

- name: Configure Core RDP Settings

run: |

Enable Remote Desktop and disable Network Level Authentication (if needed)

Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' `

-Name "fDenyTSConnections" -Value 0 -Force

Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal

Server\WinStations\RDP-Tcp' `

-Name "UserAuthentication" -Value 0 -Force

Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal

Server\WinStations\RDP-Tcp' `

-Name "SecurityLayer" -Value 0 -Force

Remove any existing rule with the same name to avoid duplication

netsh advfirewall firewall delete rule name="RDP-Tailscale"

For testing, allow any incoming connection on port 3389

netsh advfirewall firewall add rule name="RDP-Tailscale" `

dir=in action=allow protocol=TCP localport=3389

(Optional) Restart the Remote Desktop service to ensure changes take effect

Restart-Service -Name TermService -Force

- name: Create RDP User with Secure Password

run: |

Add-Type -AssemblyName System.Security

\$charSet = @{

Upper = [char[]](65..90) # A-Z

Lower = [char[]](97..122) # a-z

Number = [char[]](48..57) # 0-9

Special = ([char[]](33..47) + [char[]](58..64) +

[char[]](91..96) + [char[]](123..126)) # Special characters

}

\$rawPassword = @()

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$rawPassword += $charSet.Upper | Get-Random -Count 4
$rawPassword += $charSet.Lower | Get-Random -Count 4
$rawPassword += $charSet.Number | Get-Random -Count 4
$rawPassword += $charSet.Special | Get-Random -Count 4
$password = -join ($rawPassword | Sort-Object { Get-Random })
$securePass = ConvertTo-SecureString $password -AsPlainText -Force
New-LocalUser -Name "RDP" -Password $securePass -AccountNeverExpires
Add-LocalGroupMember -Group "Administrators" -Member "RDP"
Add-LocalGroupMember -Group "Remote Desktop Users" -Member "RDP"

echo "RDP_CREDS=User: RDP | Password: $password" >> $env:GITHUB_ENV

if (-not (Get-LocalUser -Name "RDP")) {
    Write-Error "User creation failed"
    exit 1
}

- name: Install Tailscale
run: |
    $tsUrl = "https://pkgs.tailscale.com/stable/tailscale-setup-1.82.0-amd64.msi"
    $installerPath = "$env:TEMP\tailscale.msi"

    Invoke-WebRequest -Uri $tsUrl -OutFile $installerPath
    Start-Process msiexec.exe -ArgumentList "/i", ""$installerPath`"", "/quiet", "/norestart"
-Wait
    Remove-Item $installerPath -Force

- name: Establish Tailscale Connection
run: |
    # Bring up Tailscale with the provided auth key and set a unique hostname
    & "$env:ProgramFiles\Tailscale\tailscale.exe" up --authkey=${{
secrets.TAILSCALE_AUTH_KEY }} --hostname=gh-runner-$env:GITHUB_RUN_ID

    # Wait for Tailscale to assign an IP
    $tsIP = $null
    $retries = 0
    while (-not $tsIP -and $retries -lt 10) {
        $tsIP = & "$env:ProgramFiles\Tailscale\tailscale.exe" ip -4
        Start-Sleep -Seconds 5
        $retries++
    }

    if (-not $tsIP) {
        Write-Error "Tailscale IP not assigned. Exiting."
    }

```

```

    exit 1
}
echo "TAILSCALE_IP=${tsIP}" >> $env:GITHUB_ENV

- name: Verify RDP Accessibility
  run: |
    Write-Host "Tailscale IP: $env:TAILSCALE_IP"

    # Test connectivity using Test-NetConnection against the Tailscale IP on port 3389
    $testResult = Test-NetConnection -ComputerName $env:TAILSCALE_IP -Port 3389
    if (-not $testResult.TcpTestSucceeded) {
      Write-Error "TCP connection to RDP port 3389 failed"
      exit 1
    }
    Write-Host "TCP connectivity successful!"

- name: Maintain Connection
  run: |
    Write-Host "`n=== RDP ACCESS ==="
    Write-Host "Address: $env:TAILSCALE_IP"
    Write-Host "Username: RDP"
    Write-Host "Password: $(echo $env:RDP_CREDS)"
    Write-Host "=====`n"

    # Keep runner active indefinitely (or until manually cancelled)
    while ($true) {
      Write-Host "[$(Get-Date)] RDP Active - Use Ctrl+C in workflow to terminate"
      Start-Sleep -Seconds 300
    }

```